
Test-Time Refinement Moves Intelligibility Six Times More Than Identity in Masked-Diffusion TTS

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 A diffusion language model buys sequential computation twice: depth, paid once
2 in parameters, and denoising steps, rented per utterance at serve time. We ask
3 whether the rented kind helps every capability equally. We train an iso-parameter
4 width–depth grid of 15 masked-diffusion codec TTS models (3 seeds, 20–125 M
5 non-embedding parameters) and sweep the refinement budget T at inference only,
6 giving a 75-point (w, d, T) surface per metric from 45 trained models. Fitting
7 $\text{err} = E + Aw^{-\alpha} + Bd^{-\beta} + CT^{-\tau}$ pre-registered, test-time scaling is metric-
8 selective: $\Delta\tau = \tau_{\text{WER}} - \tau_{\text{SIM}} = 0.110$, 95% run-level bootstrap CI [0.092, 0.131],
9 excluding zero. The effect that carries practical weight is one of *magnitude*: over
10 $T=1 \rightarrow 16$ refinement moves 0.952 absolute WER (a $6.1\times$ reduction) against
11 0.161 absolute speaker-similarity error ($1.25\times$), a $5.9\times$ asymmetry. It is not a
12 difference in convergence *rate*: on each metric’s own normalised curve the fraction
13 of total gain realised by $T=8$ is 94.6% for WER and 93.7% for identity, identity
14 marginally later. The selectivity persists across a 276 M four-budget replication,
15 $\Delta\tau = +0.1095$, CI [0.0951, 0.1241]. The asymmetry is less a limit on what
16 inference can buy than on what *refinement* can buy: at matched NFE, best-of-
17 K search moves speaker similarity by +0.0444 (CI [0.0416, 0.0470]) against
18 refinement at $T=64$, or +0.0365 against the deployable $T=16$ default, where
19 refinement moves it least. Depth and steps are *not* interchangeable: the substitution
20 form $B(dT^\kappa)^{-\beta}$ is worse than the separable form by $\Delta\text{AICc} = +69.3$.

21 1 Introduction

22 A diffusion language model can buy sequential computation twice. It *owns* depth: d transformer
23 blocks, paid once in parameters and then on every forward pass forever. It *rents* refinement steps:
24 T denoising passes over the same sequence, paid per utterance and adjustable at serve time. An
25 autoregressive model has only the first lever; its serial cost per token is fixed by the architecture the
26 moment training ends. If rented serial compute substitutes for owned serial compute, a deployment
27 can trade parameters for steps after the fact — but only for the capabilities that steps actually improve.

28 Zero-shot text-to-speech is a clean probe: intelligibility is compositional (the right phonetic content in
29 the right order, read out by word error rate) while speaker identity is representational (timbre carried
30 from a short prompt, read out by SIM-o). The two need not respond to the same knob.

31 We train one iso-parameter width–depth grid of masked-diffusion codec language models and then
32 sweep T at inference only, which yields a full (w, d, T) response surface per metric at zero additional
33 training cost. Two claims were pre-registered before any run. **H-D2**: refinement steps improve
34 intelligibility more than identity, i.e. test-time scaling is metric-selective. **H-D3**: for intelligibility,
35 steps and depth exchange at a measurable rate κ — how many step-doublings buy one depth-doubling.

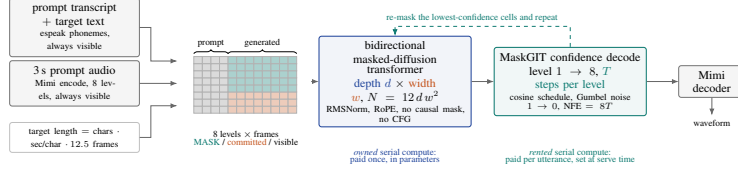


Figure 1: *The two ways a masked-diffusion TTS model buys sequential computation.* Phonemes and a 3-second prompt are always visible; the generated region of the $8 \times \text{frames}$ token grid starts fully **masked**. The backbone spends **depth** — *owned* serial compute, fixed at training time — and the sampler spends **T refinement steps per codebook level** — *rented* serial compute, chosen at serve time. One training grid over (w, d) therefore yields a whole (w, d, T) surface at inference. This paper asks whether the rented kind helps every capability equally.

$$\begin{aligned} \text{separable: } \text{err}_m(w, d, T) &= E_m + A_m w^{-\alpha_m} + B_m d^{-\beta_m} + C_m T^{-\tau_m} \\ \text{substitution: } \text{err}_m(w, d, T) &= E_m + A_m w^{-\alpha_m} + B_m (d T^{\kappa_m})^{-\beta_m} \end{aligned} \quad (1)$$

36 The first answer is yes and quantitative: $\Delta\tau = 0.110$, run-level bootstrap CI [0.092, 0.131]. The
 37 second is a clean no — the substitution form is decisively *worse* than the separable one ($\Delta\text{AICc} =$
 38 $+69.3$ for WER, $+71.3$ for SIM-o) and κ runs to its bound, so no finite depth-step exchange
 39 rate describes this regime. Extra inference compute is a targeted instrument, not a substitute for
 40 architecture.

- 41 ❶ The first per-metric (w, d, T) scaling surface for speech generation: 75 points per metric from 45
 42 models.
- 43 ❷ $\Delta\tau = 0.110$, CI [0.092, 0.131] — steps are metric-selective — with depth-step substitution
 44 reported as a *negative* AICc result ($+69.3$) and κ as unidentified.
- 45 ❸ An identity ledger against a measured codec ceiling: what moves speaker similarity when
 46 refinement does not — training compute and test-time selection.

47 2 Related work

48 Masked diffusion language models are trained at scale [Nie et al., 2025, Sahoo et al., 2024, Austin
 49 et al., 2021], but their scaling is reported against a single likelihood-shaped objective, so a per-
 50 capability question cannot arise. The masked-generative recipes that made codec-token TTS practical
 51 [Borsos et al., 2023, Chang et al., 2022, Wang et al., 2024] fixed a step budget by engineering
 52 judgement and report a single operating point; none maps error against steps as a law. Autoregressive
 53 codec TTS [Wang et al., 2023] reports scaling in total parameters only [Ye et al., 2025], which
 54 separates neither width from depth nor either from steps, as is true of classical scaling work [Kaplan
 55 et al., 2020, Hoffmann et al., 2022]. Test-time scaling is well studied in reasoning [Snell et al.,
 56 2024], where the payoff is one correctness metric; whether extra inference compute helps different
 57 capabilities *differently* is, to our knowledge, untested in generative speech.

58 3 Method

59 **Backbone and objective.** A bidirectional (non-causal) pre-LN transformer over Mimi [Défossez
 60 et al., 2024] codec tokens at 12.5 Hz with the first 8 residual codebooks (2048 entries each, plus
 61 MASK and PAD). **Width** w , **depth** d , head dimension 64, GELU MLP ratio 4, RMSNorm, RoPE,
 62 no weight tying, so the non-embedding parameter count is $N = 12 d w^2$. One embedding table
 63 per codebook is summed at the input and one linear head per codebook reads out. Training is
 64 SoundStorm-style [Borsos et al., 2023] coarse-to-fine masking: per example a level $l \sim \mathcal{U}\{1..8\}$
 65 and a ratio $t \sim \mathcal{U}[0, 1]$; levels below l stay visible, level l is masked i.i.d. with probability t , levels
 66 above l are entirely MASK; the loss is cross-entropy on the masked cells of level l only. A phoneme
 67 prefix and a 3-second prompt of codec tokens are always visible; there is no speaker encoder, so the
 68 backbone is the only identity pathway, and no classifier-free guidance, which would confound the
 69 step axis. The target length is fixed before generation from the text length and a corpus character rate
 70 (0.0602 s/char).

Frozen sampler. MaskGIT-style confidence decoding [Chang et al., 2022], level by level, with exactly T refinement steps per level, so $\text{NFE} = 8T$: a cosine schedule sets how many cells remain masked after each step, tokens are drawn at temperature 1.0, and confidences are log-probabilities plus Gumbel noise annealed $1 \rightarrow 0$; committed cells stay committed. The sampler is identical for every config, seed and T , and generation noise is keyed so that T is the only difference between surface points of one run. We verify T reaches the sampler before computing any metric: $T=1$ and $T=16$ must differ on $>20\%$ of cells.

Design and statistics. 15 configs span three iso- N budgets $\times$ five aspect ratios, each trained at 3 seeds for 30,000 steps at effective batch 256 with μP [Yang et al., 2021] (base width 256, base LR 0.004; transfer verified by five-point LR sweeps). $T \in \{1, 2, 4, 8, 16\}$ is swept at evaluation only, giving 75 surface points per metric from 45 models. Metrics: WER (Whisper-large-v3 [Radford et al., 2023], greedy), SIM-o (WavLM-large [Chen et al., 2022] against the *original* prompt waveform, following Anastassiou et al. [2024]), a frozen degenerate-output rule, and UTMOS [Saeki et al., 2022]. Error transforms are $\text{err}_{\text{WER}} = \text{WER}$ and $\text{err}_{\text{SIM}} = 1 - \text{SIM-o}$, computed over all items including degenerates. Both forms of Eq. (1) are fitted by weighted non-linear least squares with 32 multi-starts; $\Delta\tau = \tau_{\text{WER}} - \tau_{\text{SIM}}$, κ , and $\Delta\rho = \alpha_{\text{SIM}}/\beta_{\text{SIM}} - \alpha_{\text{WER}}/\beta_{\text{WER}}$ carry percentile confidence intervals from a bootstrap that resamples at the *run* level — the five T -points of one run share a trained model, so resampling surface points would understate their correlation — with 2000 replicates recomputing every T point.

4 Results

Steps are metric-selective (H-D2, supported). Fitted on all 75 surface points, the step exponent is $\tau_{\text{WER}} = 0.836$ (CI [0.825, 0.851]) against $\tau_{\text{SIM}} = 0.726$ (CI [0.709, 0.744]), so $\Delta\tau = 0.110$ with CI [0.092, 0.131], excluding zero. The raw surface shows it without any model: over the grid, $T=1 \rightarrow T=16$ takes WER from 113.8% to 18.7% (a factor of $6.1\times$ on the error scale) while SIM-o moves only $0.209 \rightarrow 0.370$ ($1.25\times$ on $1 - \text{SIM-o}$), an absolute-error asymmetry of $5.9\times$. The selectivity is not an artefact of scale, training compute, or the metric stack: a fourth budget (276M, ten further runs) gives $\Delta\tau = +0.1095$, CI [0.0951, 0.1241]; retraining three configurations $3\times$ longer gives $+0.1767$, CI [0.0789, 0.2854]; and a different ASR or unbounded log-amplitudes move it to $+0.1283$ and $+0.1260$. Extending T to 64 shows WER still falling past the grid edge ($0.1516 \rightarrow 0.1294$), so the $T=16$ value was a lower bound. The power gate passed in all three budgets before these tests ran.

Depth and steps do not substitute (H-D3, not supported). The pre-registered substitution form collapses: $\Delta\text{AICc}(M_{\text{sub}} - M_{\text{sep}}) = +69.3$ for WER and $+71.3$ for SIM-o where the hypothesis required $\leq +4$, and κ sits on its bound for both metrics, so we quote none. The fits show why: T enters as an additive term decaying to a floor, while depth moves the floor itself, and one factor dT^κ cannot fit both orderings.

Shape matters, but the anisotropy ratio is not identified (H-D1, not supported). At $T=16$ the two-axis model beats the N -only model for both metrics ($\Delta\text{AICc} = -15.5 / -5.5$ against a -4 threshold), so error is not a function of parameter count alone. The *direction* of the anisotropy we cannot claim: $\Delta\rho = 0.174$, CI [-0.711, 0.388] spanning zero, because the width amplitude saturates its pre-registered bound in every fit, leaving only $A w^{-\alpha}$ identified. *Depth* is identified and separates the capabilities sharply ($\beta_{\text{WER}} = 2.040$ versus $\beta_{\text{SIM}} = 0.553$): moving from the shallowest shape to the best depth is worth 40.2, 16.3 and 7.3 WER points across the three budgets, while SIM-o varies by under 0.065. The depth optimum is interior only in the smallest budget; in the two larger ones the argmin sits at the deepest shape tested, so we claim no $d^*(N)$ trend.

Extrapolation and reliability. Fitting the two smaller budgets to predict the largest gives MAPE 2.4% for SIM-o (versus 2.2% N -only), inside the 15% asked for; WER does not extrapolate (20.2% versus 20.4%). Degeneracy falls monotonically with T , 40.4% to 0.1%: most of what $T=1$ costs is outright failure.

Serving corollary. Measured at batch 1, c_{layer} is flat in width (0.493–0.514 ms per layer for $w \in [256, 1152]$): the device is launch-bound, so serial latency is $8Td c_{\text{layer}}$ and width is free,

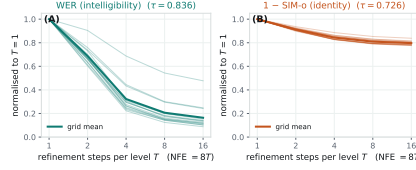


Figure 2: *Test-time scaling is metric-selective.* Error versus refinement steps for all 15 configs, each normalised to its own $T=1$ value; thick lines are grid means. (A) WER error falls 0.9518 in absolute terms over $T=1 \rightarrow 16$ and is still falling at $T=16$. (B) SIM-o error falls 0.1606 over the same range, $5.9\times$ less. Same models, same frozen sampler, same items — only T differs.

Table 1: *Fitted exponents* on the 75-point surface (M_{sep} for α, β, τ ; M_{sub} for κ) with 95% run-level bootstrap CIs for the step exponents. $\Delta\tau = 0.110$, CI [0.092, 0.131], excludes zero: the pre-registered primary result. κ is at its bound in both rows and $\Delta\text{AIC}_{\text{csub-sep}} \gg +4$, so no exchange rate is quotable. $\Delta\text{AIC}_{\text{cfull-N}} \leq -4$ means iso- N shape matters.

Metric	α (width)	β (depth)	τ (steps)	κ	$\Delta\text{AIC}_{\text{csub-sep}}$	$\Delta\text{AIC}_{\text{cfull-N}}$
WER	1.574	2.040	0.836 [0.825, 0.851]	3.000 [†]	+69.3	−15.5
SIM-o	0.919	0.553	0.726 [0.709, 0.744]	3.000 [†]	+71.3	−5.5

[†]at the fitted bound, i.e. not identified; $\Delta\tau = 0.110$, CI [0.092, 0.131].

122 leaving a latency budget to constrain only the product Td . Our data do not resolve how to split it —
 123 the pre-registered depth-before-steps rule held at 5.9% of 40 budgets against an 80% bar, and T is
 124 censored at 16 — so we claim no directional allocation rule.

125 **What buys identity, measured against a ceiling.** If refinement is an intelligibility knob, what
 126 moves identity? We measure the candidates on one scale against a bound: a codec round-trip of
 127 ground-truth audio scores 0.555, leaving 0.0747 of headroom above our best system. The largest lever
 128 is *training compute* (30k→90k steps, +0.0827, CI [0.0733, 0.0908] over 9 runs; a further doubling
 129 adds +0.0155). Next is *test-time search*: at matched NFE ($8T$ per candidate), best-of- K sampling
 130 with candidates chosen by one speaker encoder and scored by a second, independently gated one
 131 (§ Limitations) improves similarity by +0.0444 (CI [0.0416, 0.0470], 72 % of items) over refinement
 132 at $T=64$, or +0.0365 over the deployable $T=16$ default for +0.0425 WER — while refinement
 133 stays the better spend for intelligibility (WER 0.1316 versus 0.1955). The gain is selection, not
 134 sampling: an arbitrary candidate scores 0.4259 regardless of K . It persists at 276 M non-embedding
 135 parameters (+0.0422) and, on three runs, at $3\times$ training compute (+0.0368). Further behind sit a $14\times$
 136 parameter increase (+0.0531) and training-free speaker-contrastive guidance (+0.0121, for +2.73
 137 WER points — a trade), while rate-matched length conditioning (+0.0044) and longer prompts buy
 138 nothing: the best prompt is the 3 s default seen in training, and beyond it output stays fluent and
 139 speaker-consistent while decoupling from the text. Identity is bought with training and selection, not
 140 with more refinement.

141 *Where* the passes are spent matters as much as how many: at matched NFE = 32, moving them to the
 142 coarsest codebook level changes WER from 1.1480 to 0.2191, at a cost in identity and naturalness
 143 (Appendix D).

144 5 Limitations and outlook

145 Our ceiling is 133 M non-embedding parameters and 30,000 steps, so every model is under-trained
 146 and T_{WER}^* is a lower bound. Part of the identity gap is not ours to close: a codec round-trip of
 147 the target scores 0.555 against 0.678 for the original, so 62% of the distance to real speech is lost
 148 in the codec. The search result carries one limitation we could not close: selector and scorer are
 149 both ECAPA-TDNN-family encoders (item-level $r = 0.71 - 0.73$); each passes the ground-truth
 150 speaker gate independently and the selector captures only 46 % of an oracle’s gap where shared bias
 151 would predict nearly all of it, but excluding representational overlap needs a different scorer family,
 152 and matching is on diffusion NFE only. One language, one codec, one corpus, and no classifier-free
 153 guidance. The $\Delta\tau$ interval captures seed variation only; target length comes from a corpus median;
 154 $\Delta\rho$ and κ are unidentified. Cuts taken: none; pre-registered outcome class S1.

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209 A Reproducibility

210 Protocol, hypotheses, thresholds, the frozen sampler and the analysis plan were registered before the
 211 first run, and every gate is logged with its measured values. The released artefact holds the harness,
 212 all 45 run records, the 75-point surface, and the fitting and bootstrap code. Total cost 143 GPU-hours.

213 B The shape grid, as run

214 Every config in the active grid with its measured $T=16$ means (seeds 0, 1, 2, 400 eval_zs items each).
 215 Iso- N within a budget: the three budgets sit at 20, 50 and 125 M non-embedding parameters and the
 216 five shapes per budget span the aspect-ratio range at fixed $N = 12 d w^2$.

	config	budget	w	d	heads	N (M)	val loss	WER (%)	SIM-o	Degen (%)
	A1	A	640	4	10	19.7	4.638	53.9	0.327	0.67
	A2	A	448	8	7	19.3	4.582	28.4	0.348	0.08
	A3	A	384	12	6	21.2	4.566	15.9	0.360	0.17
	A4	A	320	18	5	22.1	4.574	13.6	0.359	0.00
	A5	A	256	24	4	18.9	4.611	15.3	0.351	0.17
	B1	B	832	6	13	49.8	4.551	27.8	0.361	0.08
217	B2	B	640	10	10	49.2	4.505	16.0	0.381	0.00
	B3	B	512	16	8	50.3	4.497	11.9	0.388	0.08
	B4	B	448	22	7	53.0	4.511	12.0	0.382	0.08
	B5	B	384	30	6	53.1	4.520	11.5	0.375	0.08
	C1	C	1152	8	18	127.4	4.504	17.1	0.373	0.17
	C2	C	960	12	15	132.7	4.504	18.9	0.373	0.17
	C3	C	768	18	12	127.4	4.488	14.3	0.392	0.00
	C4	C	640	26	10	127.8	4.489	13.4	0.390	0.08
	C5	C	512	36	8	113.2	4.487	9.9	0.388	0.00

218 C Additional figures

219 D Refinement-budget allocation across codebook levels

220 Three frozen per-level step schedules at matched total NFE = 32 on the 125 M budget (15 runs, 400
 221 items). Paired item-level bootstrap, 2000 replicates, pairing on the eval item so item difficulty cancels.
 222 Coarse spends all passes on codebook level 0, fine on level 7, uniform splits evenly.

	schedule	WER	SIM-o	UTMOS	degenerate
223	coarse [25, 1, ..., 1]	0.2191	0.3057	2.17	0.1%
	uniform [4, ..., 4]	0.3144	0.3477	2.51	0.5%
	fine [1, ..., 1, 25]	1.1480	0.2246	1.56	39.1%

224 Allocation is metric-selective in the same way the step *count* is: coarse wins WER but loses identity
 225 and naturalness against uniform, both with intervals excluding zero. It is a trade, not a free win.

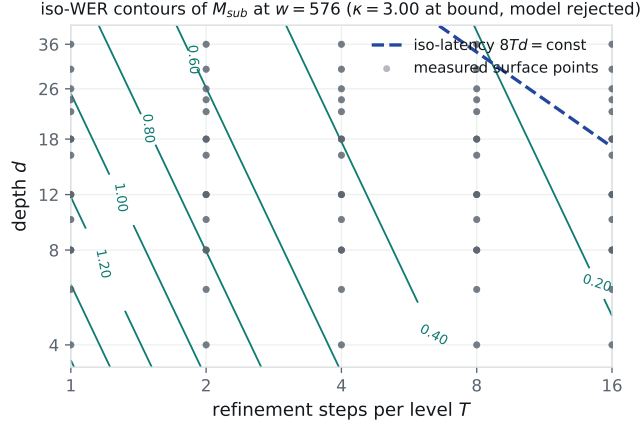


Figure 3: *Why the substitution form is rejected.* Iso-WER contours of M_{sub} in $(\log T, \log d)$ at the median width, with the iso-latency line $8Td = \text{const}$. The fitted κ sits on its bound and $\Delta\text{AICc}(M_{\text{sub}} - M_{\text{sep}}) = +69.3$, so these contours are shown as a rejected model, not as a design rule.

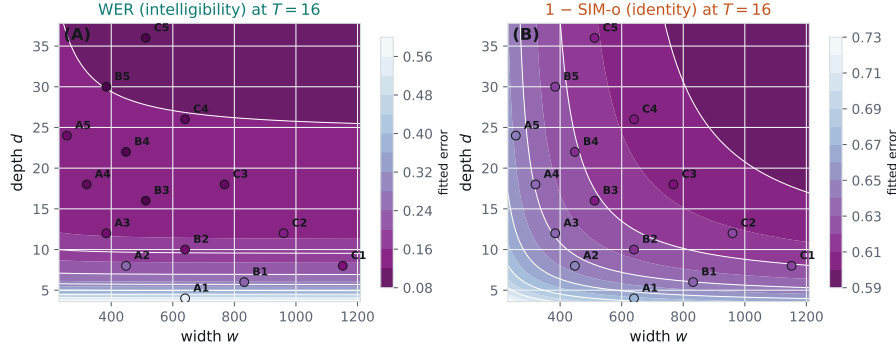


Figure 4: *Fitted anisotropy surface at $T=16$ for both metrics, with the measured config means overlaid.* The width amplitude A saturates its pre-registered upper bound for both metrics, which is why $\rho = \alpha/\beta$ is reported as unidentified in §4 even though the two-axis model beats the N -only model ($\Delta\text{AICc} = -15.5$ and -5.5).

226 **A significance-based saturation contrast.** Defining saturation as the smallest T whose paired-
 227 bootstrap interval for $\text{err}(T) - \text{err}(64)$ covers zero gives $T_{\text{WER}}^* = 32$ against $T_{\text{SIM}}^* = 16$ on the
 228 extended grid. We report this only here and do not build on it: roughly half the contrast is a precision
 229 artefact, since identity’s detection floor is $\approx 1.8\times$ looser relative to its own dynamic range than
 230 intelligibility’s. The raw-scale T^* — 95% of the $T=16$ value — is not affine-invariant and must
 231 never be compared across metrics: the same 5% band is 0.67% of WER’s range but 11.96% of
 232 SIM-o’s. On the invariant fraction-of-total-gain definition, $T^* = 16$ for both metrics.

233 E Fit diagnostics and exploratory analyses

234 **Parameters at bounds.** The width amplitude A saturates the pre-registered *upper* bound $A \leq 10$ in
 235 every fit (Part A: 10.000 for both metrics; Part B: 9.999 for WER, 10.000 for SIM-o), so only the
 236 product $A w^{-\alpha}$ is identified and α is not; the bound, not the data, is binding, which is a limitation
 237 of the pre-registered parameterisation worth fixing in any follow-up. κ reaches its upper bound in
 238 every one of the 2000 bootstrap replicates of M_{sub} . The irreducible term E reaches 0 for WER under
 239 M_{sep} . The depth amplitude B is interior throughout, so β is identified.

240 **Exploratory: the depth exponent separates the capabilities.** $\beta_{\text{WER}} = 2.040$ against $\beta_{\text{SIM}} =$
 241 0.553 , a factor of 3.7 — the asymmetry H-D1 sought, in the parameter the data pin down. Part-B ρ

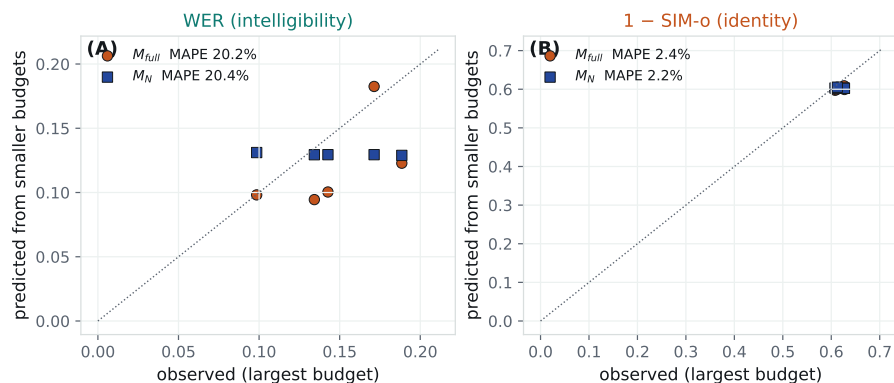


Figure 5: *H-D4 extrapolation*. The two smaller budgets are fitted and the largest budget predicted, under M_{full} and the N -only model. SIM-o extrapolates (MAPE 2.4 % vs 2.2 %); WER does not (20.2 % vs 20.4 %).

values point the pre-registered way ($\Delta\rho = +0.890$) but inherit the A -cap problem; the pre-registered test is Part A, so this is exploratory only.

Weighting. Points are weighted $1/SE^2$ with SE taken over seeds, floored at a quarter of the pooled seed SD so that a chance agreement between two seeds cannot dominate the fit; the number of floored points is recorded in the released `fits.json`.

Exploratory (not part of any declared claim). An unweighted refit of both parts is released alongside the weighted fit, as is a fit that additionally includes the finished runs of a stretch third seed. Both reproduce the sign and significance of $\Delta\tau$.

Use of Generative AI Disclosure

Generative AI was used as a coding assistant for the experimental harness and for language refinement of author-written text. The research design, the pre-registered hypotheses and thresholds, the analysis plan, and all scientific claims are the authors' own; the authors verified the code and the numbers and take full responsibility for this paper.